Highland & 4И Statistics Spring 2025 HW2

Problem 1: In 1861, 10 essays appeared in the New Orleans Daily Crescent. They were signed "Quintus Curtius Snodgrass" and some people suspected they were actually written by Mark Twain. To investigate this, we will consider the proportion of three letter words found in an author's work. From eight Twain essays we have:

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T = [0.225, 0.262, 0.217, 0.240, 0.230, 0.229, 0.235, 0.217]
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From 10 Snodgrass essays we have:

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S = [0.209, 0.205, 0.196, 0.210, 0.202, 0.207, 0.224, 0.223, 0.220, 0.201]
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1. Perform a Wald test for equality of the means. Report the p-value and a 95% confidence interval for the difference of means. What do you conclude?
2. Now use a permutation test to avoid the use of large sample methods.

Problem 2: A randomized, double-blind experiment was conducted to assess the effectiveness of several drugs for reducing postoperative nausea. The data are as follows.

	# of patients	incidence of nausea
Placebo	80	45
Chlorpromazine	75	26
Dimenhydrinate	85	52
Pentobarbital (100 mg)	67	35
Pentobarbital (150 mg)	85	37

1. Test each drug versus placebo at 5% α-level: for each drug, consider the 2x2 contingency table (vs placebo) and test the odds ratio for being equal to 1 (see lecture 9). Report the odds ratios and the p-values.
2. Use the Bonferroni and the FDR (Benjamini-Hochberg) method to adjust for multiple testing. What do you conclude?